

Lecture 9 Text Entry on Mobile Devices - 1

Xiaojun Bi

Stony Brook University

xiaojun@cs.stonybrook.edu

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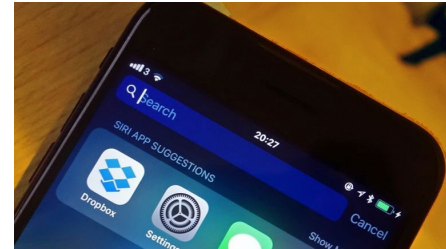
Outline

- Smart Touch Keyboard
- Gesture Typing

Text entry is a major activity in mobile interaction.



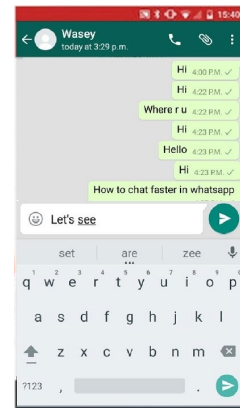
Emailing



Searching

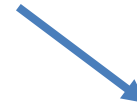
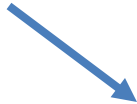


Social Networking



Messaging

Keyboard





•**Top Three Activities on Mobile Devices:**

1) Emailing, 2) Social Networking, 3) Messaging

(www.time.com)

•**18 to 24 Year Olds Average 110 Text Messages per Day**

(www.time.com)

Challenges of Touchscreen Text Entry



Challenges of Touchscreen Text Entry



Determined by key boundaries, 50% of words are not correctly typed on phone.

Unsuccessful Auto-correction

Your mom and I are going
to divorce next month

what??? why! call me
please?

I wrote Disney and this
phone changed it. We are
going to Disney.

DAMN YOU AUTOCORRECT.COM

Unsuccessful Auto-correction



Outline

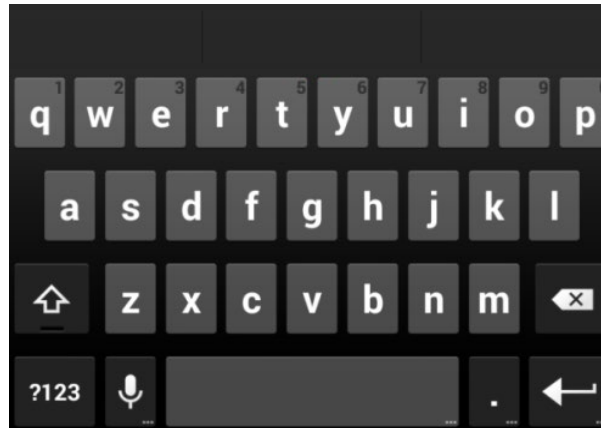
- Smart Touch Keyboard
- Gesture Typing

Smart Touch Keyboard

Typed Word

Keyboard Output

agsim



again

quivj



quick

fav

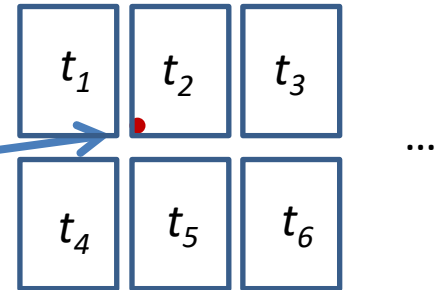


favorite

Principles of Target Selection

n target candidates: $T = \{t_1, t_2, \dots, t_n\}$

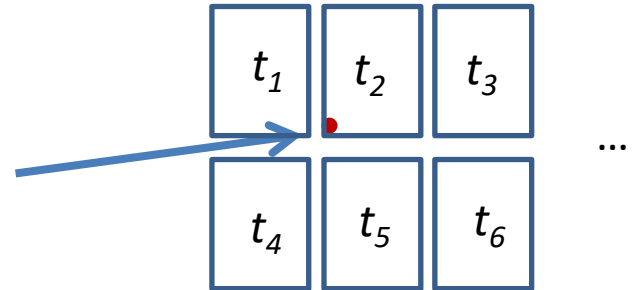
touch point: s



Principles of Target Selection

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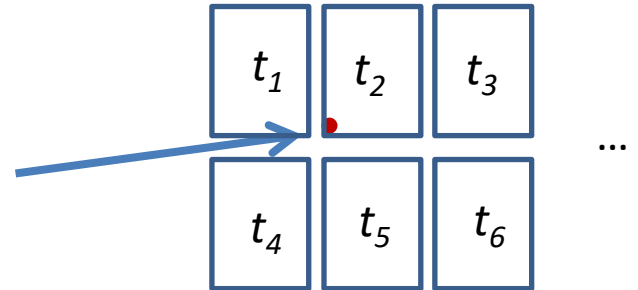
Determining the target:

$$t^* = \underset{t}{\operatorname{argmax}} P(t|s)$$

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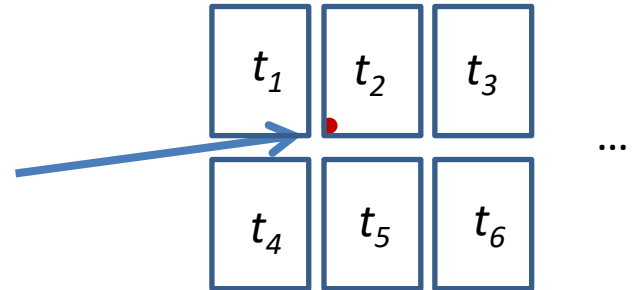
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From Bayes' rule, $P(t|s) = \frac{P(s|t)P(t)}{P(s)}$

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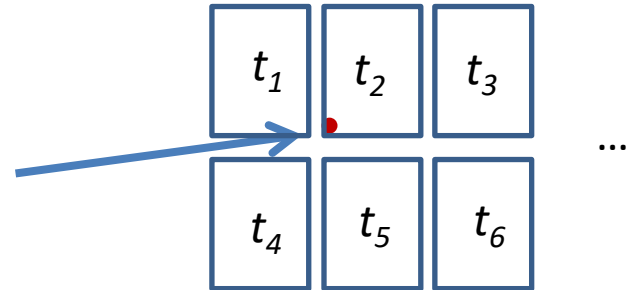
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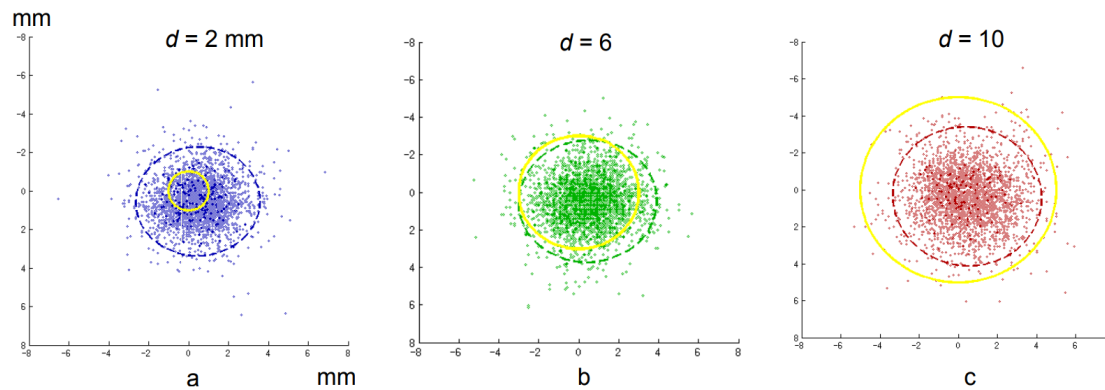
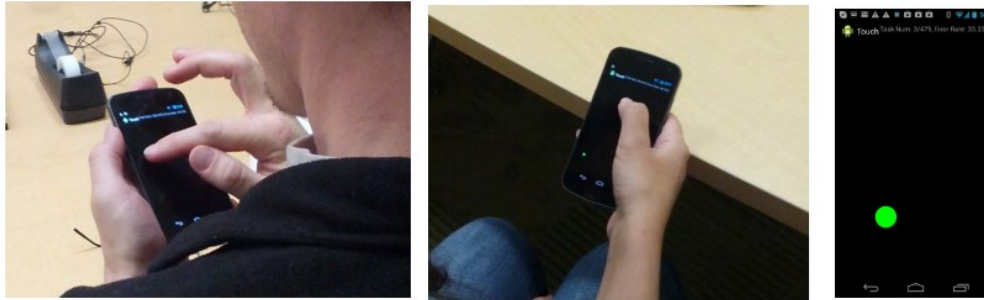
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$P(t)$: prior probability

Understanding Touch



$$S \sim N(\mu, \Sigma)$$
$$\mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}$$
$$\Sigma = \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{pmatrix}$$

Understanding Touch

Estimate variances in x and y directions

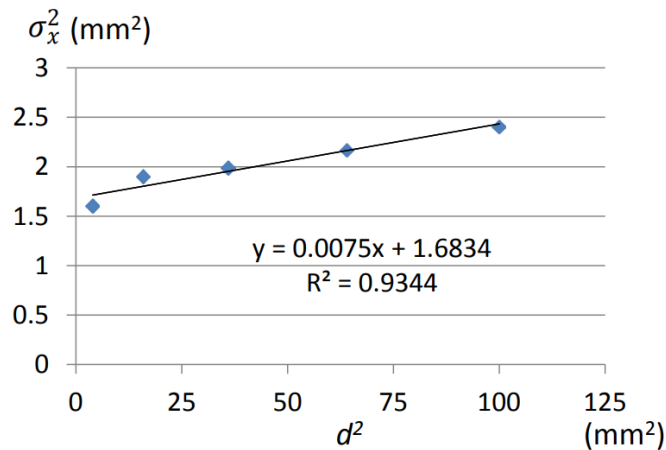


Figure 2. Regression between variance on X direction (σ_x^2) and target size (d^2).

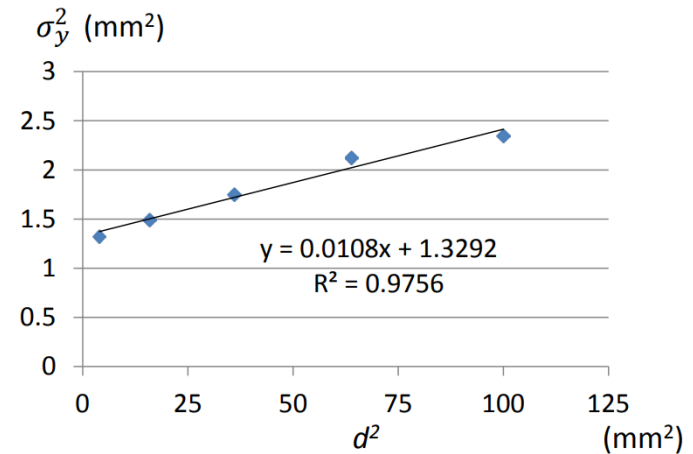


Figure 3. Regression between variance on Y direction (σ_y^2) and target size (d^2).

$$\sigma^2 = \sigma_r^2 + \sigma_\alpha^2 = \alpha \times d^2 + \sigma_\alpha^2,$$

$$P(s|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(s - \mu)^2\right\}$$

Measures for Acquiring 2-Dimensional Circular Targets

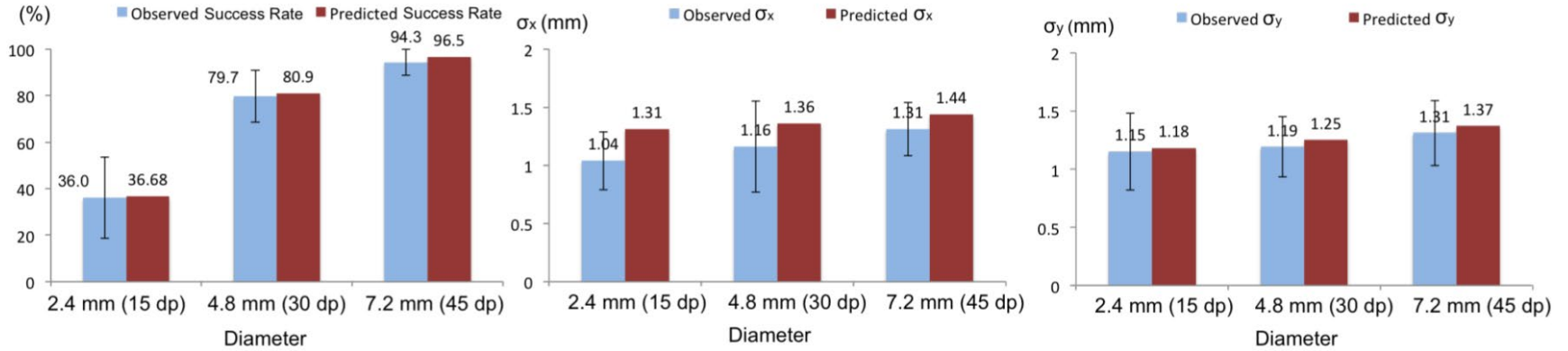


Figure 4. Predicted vs. Observed Measures in 2-Dimensional Circular Target Acquisition Tasks. Left: Success Rate, Middle: Standard Deviation of Touch Point Distribution in x direction, Right: Standard Deviation of Touch Point Distribution in y direction. Error bars are Std. Dev across participants

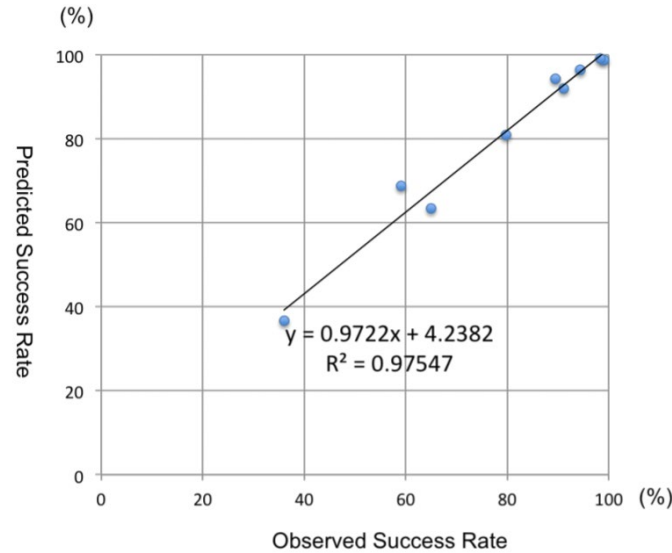
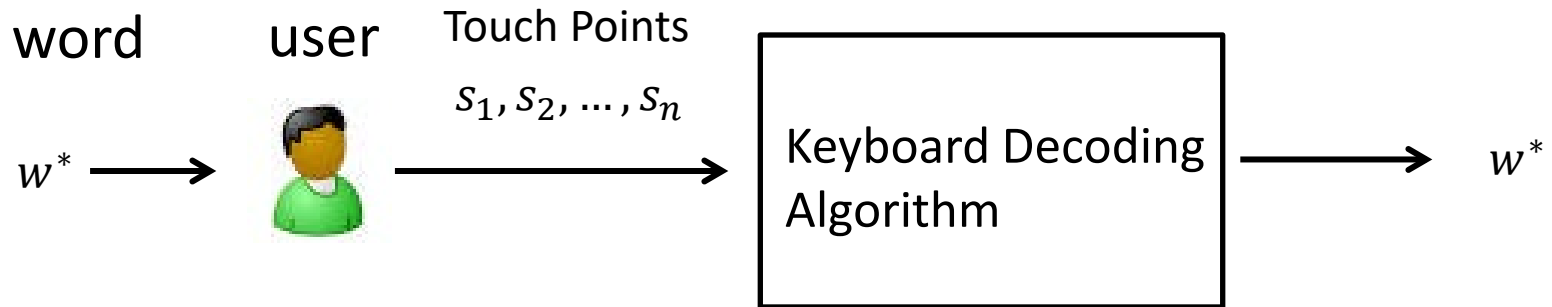
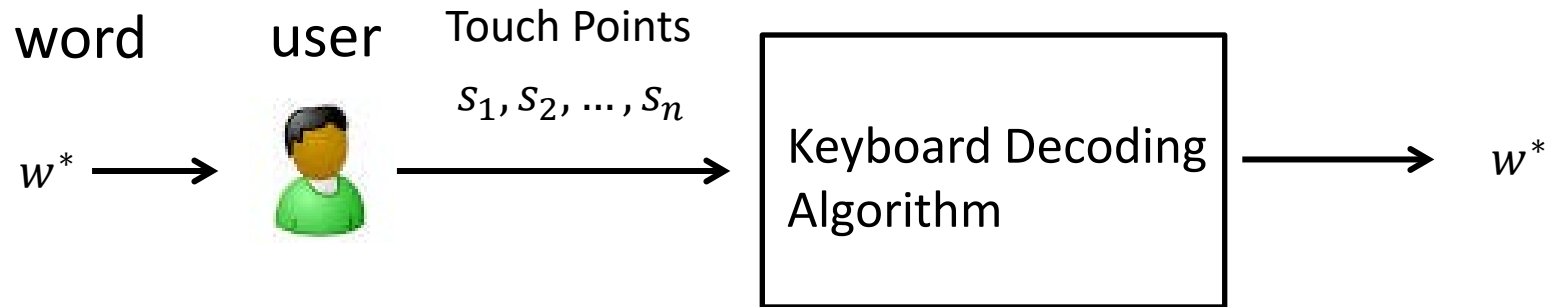


Figure 6. Predicted Success Rates vs. Observed Success Rates for each Target Size $\times$ Target Shape

Principles of Smart Touch Keyboard

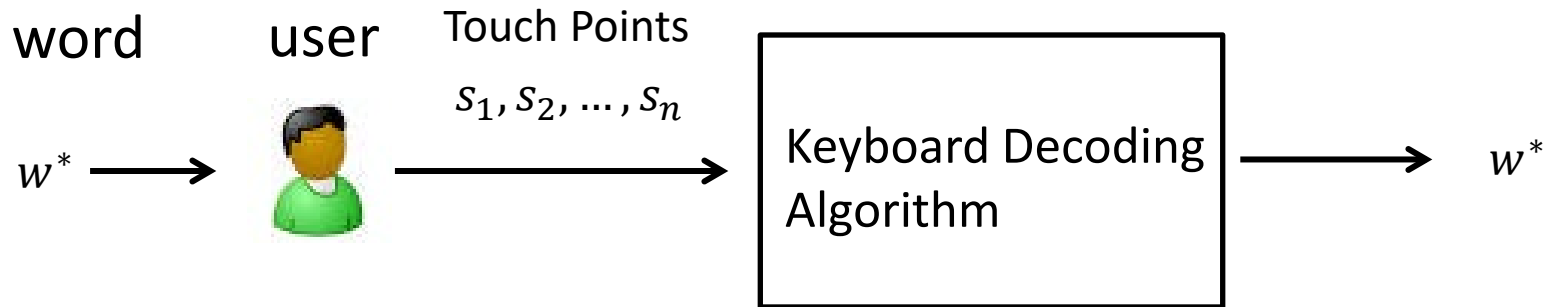


Principles of Smart Touch Keyboard



$$w^* = \underset{w}{\operatorname{argmax}} P(w | s_1, s_2, \dots, s_n)$$

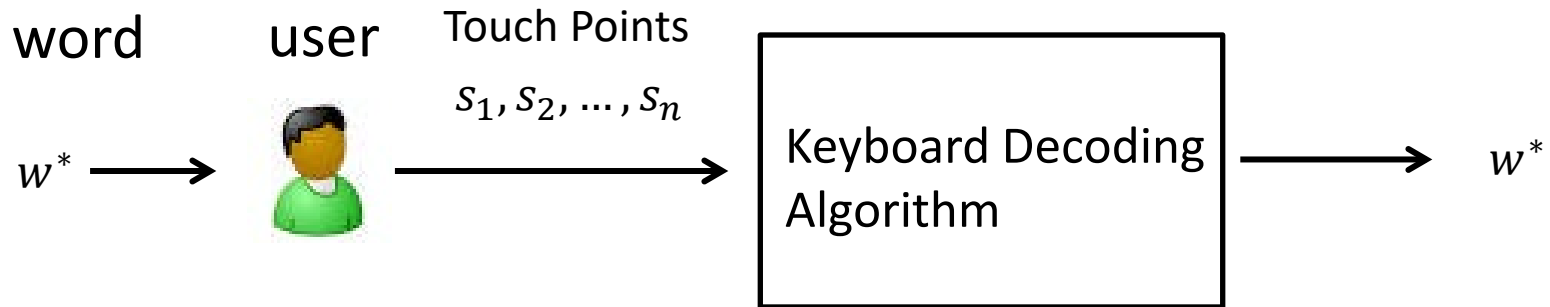
Principles of Smart Touch Keyboard



$$w^* = \underset{w}{\operatorname{argmax}} P(w | s_1, s_2, \dots, s_n)$$

$$P(w | s_1, s_2, \dots, s_n) = \frac{P(s_1, s_2, \dots, s_n | w) P(w)}{P(s_1, s_2, \dots, s_n)}$$

Principles of Smart Touch Keyboard

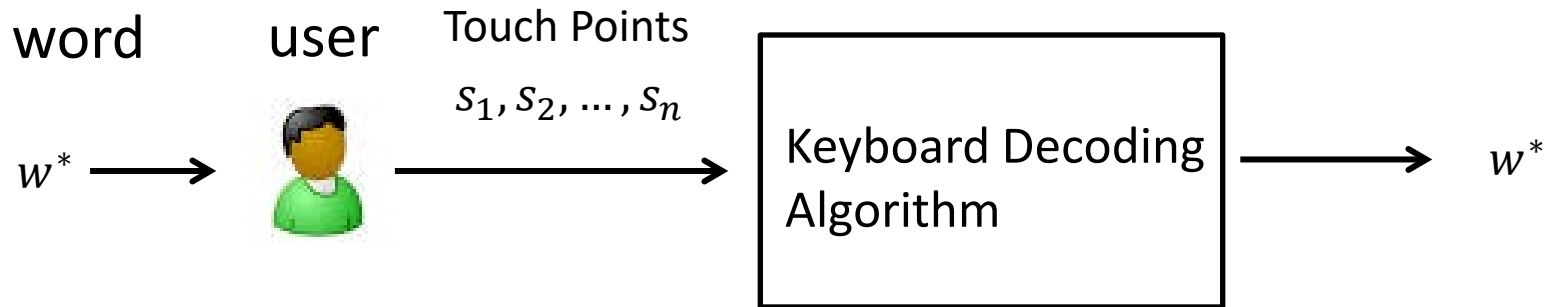


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$P(w)$: probability from language model

Principles of Smart Touch Keyboard



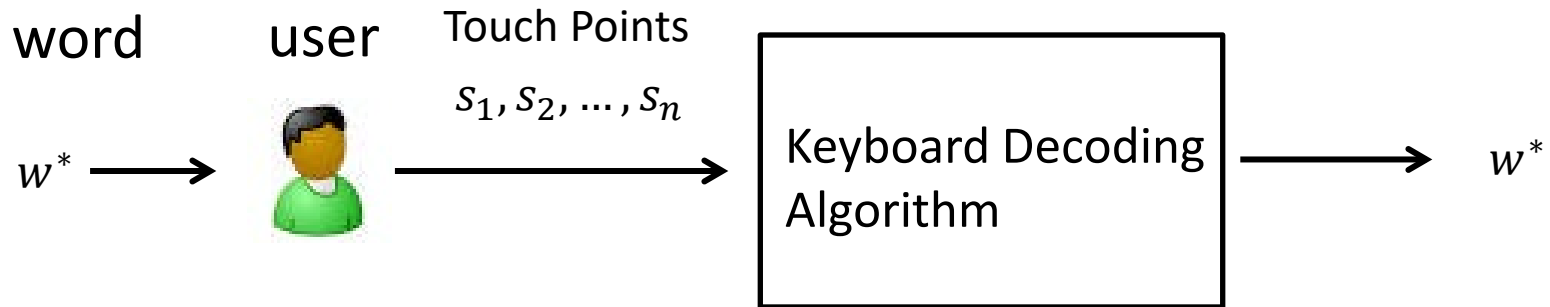
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$P(w)$: probability from language model

$P(s_1, s_2, \dots, s_n | w)$?

Principles of Smart Touch Keyboard



Assume w consists of n letters $c_1, c_2, \dots, c_n$

$$\begin{aligned} P(s_1, s_2, \dots, s_n | w) &= P(s_1, s_2, \dots, s_n | c_1, c_2, \dots, c_n) \\ &= P(s_1 | c_1) P(s_2 | c_2) \dots P(s_n | c_n) \end{aligned}$$

$$P(s_i | c_i) = P(s_i | \mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2} (s - \mu)^2\right\}$$

Question: On a smart touchscreen keyboard for language L which has three letters (A, B, C only), we have observed 4 touch points (S1, S2, S3, and S4). The following tables shows the probability of observing these touch points if a letter being the intended target.

	A	B	C
S1	0.6	0.3	0.1

	A	B	C
S2	0.4	0.3	0.3

	A	B	C
S3	0.1	0.5	0.4

	A	B	C
S4.	0.8	0.1	0.1

1. Please calculate the probability of the following words, given these four touch points. We assume the prior probability of each word is identical.

“ABCC” “BCAA” “AACB”

“ABCC”: $0.6 \times 0.3 \times 0.4 \times 0.1 = 0.0072$

“BCAA”: $0.3 \times 0.3 \times 0.1 \times 0.8 = 0.0072$

“AACB”: $0.6 \times 0.4 \times 0.4 \times 0.1 = 0.0096$

1. Please calculate the probability of the following words, given these four touch points. We assume the prior probability of each word is identical.

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“ABCC”: $0.6 \times 0.3 \times 0.4 \times 0.1 = 0.0072$

“BCAA”: $0.3 \times 0.3 \times 0.1 \times 0.8 = 0.0072$

“AACB”: $0.6 \times 0.4 \times 0.4 \times 0.1 = 0.0096$

2. Given the language context, we assume that the probability of a user to enter the following three words are:

ABCC	BCAA	AACB
0.5	0.05	0.45

Taking into account the probability from language context, which word (among these three) should be the top suggestion?

Likelihood X Prior:

$$\text{ABCC: } 0.0072 \times 0.5 = 0.0036$$

$$\text{BCAA: } 0.0072 \times 0.05 = 0.00036$$

$$\text{AACB: } 0.0096 \times 0.45 = 0.00432$$

So the top suggestion is AACB